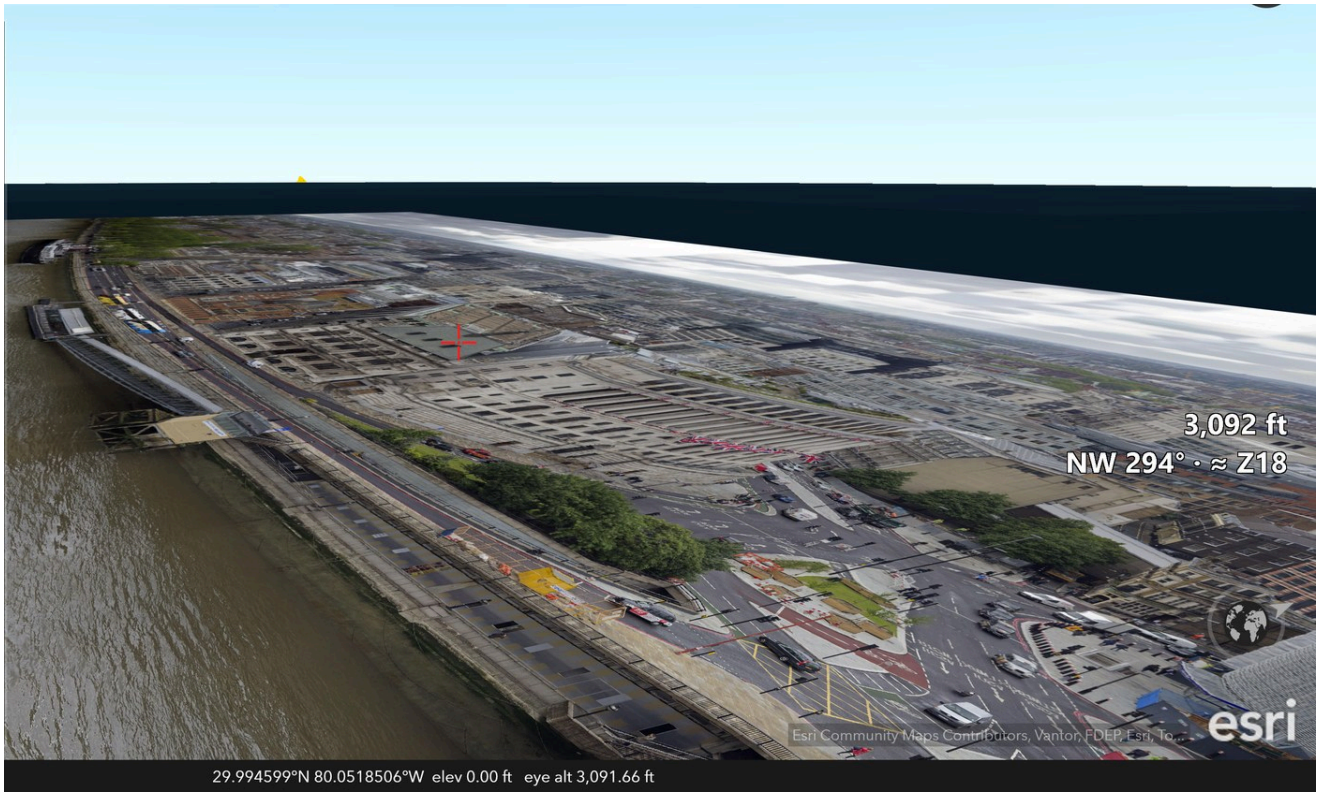
Rasta Pyramid Factory v0.2.0 - User Manual

Printable copy: Rasta_Pyramid_Factory_User_Manual.pdf

Slice. Dice. Present. Rasta Pyramid Factory turns a giant picture - a city panorama, a drone map, a scanned map, artwork - into a zoomable map you can open in **ArcGIS Earth**, then fly over and dive into. *See the whole scene like a hawk. Dive into any detail that catches your eye.*



A London panorama photo (50,000 × 12,006 pixels) made with Rasta v0.1.6 and flown in ArcGIS Earth. A photo has no real location, so Rasta parks it on the ocean off Jacksonville, Florida: the status bar shows 29.99 N, 80.05 W, elevation 0 ft. Every spot in the picture now has its own latitude and longitude, like a real map.

What you need

- **Windows 10 or 11** (64-bit).
- **Python 3.14** - free; tick "Add python.exe to PATH" if the installer shows it.
- **QGIS 3.44.9** - free; **only for real map rasters** (GeoTIFFs with coordinates), installed at the normal place, `C:\Program Files\QGIS 3.44.9`. Rasta uses it behind the scenes. **Photos and other flat pictures do not need QGIS** (photo mode, v0.2.0).
- **Pillow** - a free Python helper for photo mode; **Start Rasta Pyramid Factory.bat installs it by itself** the first time.
- **ArcGIS Earth** - free, to look at and fly over the result.

- **Memory for photo mode:** about 5 bytes per picture pixel - about 3 GB for a 600-million-pixel photo, about 6 GB for 1.2 billion.
- **Free disk space on C:** - room for the finished file twice while it builds (the working files go in the Windows temporary folder and are deleted afterwards).

Install

1. Download `Rasta_Pyramid_Factory_v0.2.0.zip`, right-click it > **Extract All**.
2. Open the `Rasta_Pyramid_Factory_v0.2.0` folder and double-click **Run Rasta Self Test.bat** once. It should end with **SELF TEST PASSED**.

What's in the zip



Build a picture map

1. For a **real map raster** (QGIS route): **close QGIS** if you have it open. Photos do not care.
2. Double-click **Start Rasta Pyramid Factory.bat**.

3. **1. Choose giant raster** - press **CHOOSE IMAGE** and pick your picture (JPG, PNG, TIF, BMP or WEBP). Rasta checks it and shows its size, whether it is a flat picture or a real map, and how many tiles it will make.
4. **2. Pyramid recipe** - leave **Min zoom 0** and **Max zoom 18** for a normal picture. Rasta suggests Max zoom 18 for flat pictures: that is where you reach the photo's own detail. The green line shows how it will be built: **PHOTO MODE - no QGIS needed** for pictures, **QGIS 3.44.9** for real map rasters.
5. **3. Finished product** - **TPKX** for ArcGIS Earth, **MBTiles** for other map programs, or **Both**. Type a **Name** and choose a **Folder**.
6. Press **BUILD RASTER PYRAMID** and leave it running. The status line keeps moving - a heartbeat, the time, and the stage (photo-mode pyramid or QGIS pyramid, converting to TPKX, verifying).
7. When it says **RASTER PYRAMID COMPLETE**, your file is in the folder you chose.

How long? In **photo mode** about **1 minute** for a 50,000 × 12,006 picture (35,216 tiles), and the TPKX is about a quarter of the size the QGIS route makes (442 MB instead of 1.77 GB for that picture). The QGIS route took 9 minutes for the same picture on Jim's PC.

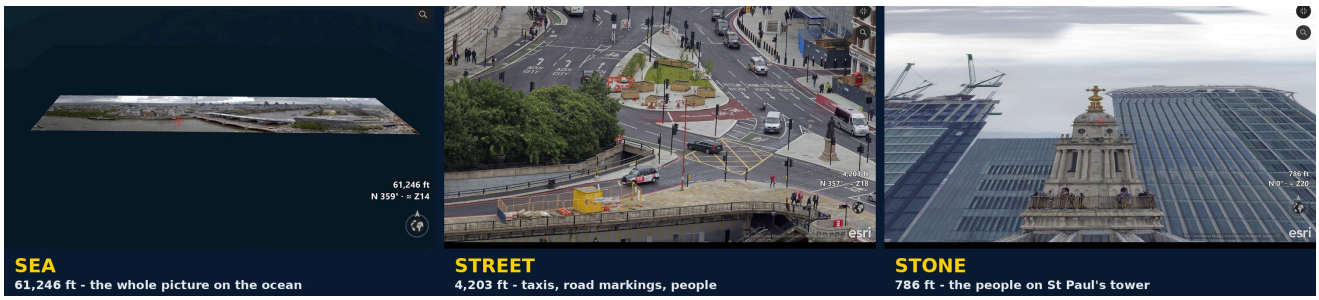
Stopping: press **CANCEL** - Rasta stops within a second or two, cleans up and says *Build cancelled*. Closing the window with the X does the same before it closes.

Look at it in ArcGIS Earth

1. Drag the `.tpkx` file onto the ArcGIS Earth map (or use the **+** Add data button).
2. If the view does not jump there, right-click the layer in the list and zoom to it.
3. **Where is it?** A real map (a GeoTIFF with coordinates) lands where it belongs. An ordinary **photo** has no location, so Rasta places it on the **Atlantic Ocean at about 30 N, 80 W**, east of Jacksonville, at **1 picture pixel = 1 map metre**. The ocean is flat and empty, so the picture lies there like a giant carpet.
4. Zoom in: the detail sharpens down to the photo's own pixels.

Fly into it with EarthStick

With EarthStick you can fly over the picture, hover above any part, and dive in. The readout's **≈ Z** number tells you when you have reached the sharpest detail (about Z18).



Sea, street, stone - one photograph, three heights: the whole panorama floating on the ocean, the street with its taxis and road markings, and the people on St Paul's bell tower.

Gates into your picture: because every spot of the picture has a latitude and longitude (hover the mouse over it and read the ArcGIS Earth status bar), you can make World Gates that drop you right onto a landmark in the photo. In World Gate Maker, type those coordinates with **ground elev 0** and a height such as 300 ft.

The picture always lies flat - it is a map layer, so it cannot stand up like a billboard. A photo taken at an angle keeps its own perspective, which gives a strange, half-3D diorama feel when you fly over it.

Messages and fixes

- **"QGIS Desktop is running"** (real map rasters only) - close QGIS, then build again.
- **"Not enough memory to open this picture in photo mode"** - close other programs, or set `engine = qgis` under `[PHOTO]` in `System Files\Rasta_Config.ini` to use the QGIS route.
- **Want all-PNG tiles (no JPEG)?** - set `tiles = png` under `[PHOTO]`. Bigger files, no JPEG compression.
- **"QGIS install folder not found" / "expects QGIS 3.44.9"** (real map rasters) - install QGIS 3.44.9 at the normal place, or put your QGIS folder in `System Files\Rasta_Config.ini` (`qgis_install_dir`).
- **"TPKX supports zoom levels up to 23"** - TPKX (Esri's tile packages) stop at zoom 23. Lower Max zoom, or choose MBTiles only (up to 25).
- **"Output already exists"** - pick another Name or move the old file.
- **It runs for a long time** - giant pictures take minutes; the moving heartbeat shows it is working. The size warning before the build tells you how many tiles to expect.
- **The disk fills up** - free space on C: (room for the finished file twice while building).

What Rasta does not do

- It cannot add detail that is not in the photo - a blurry photo stays blurry when you zoom in.
- A photo parked on the ocean is not real geography; its location is a display spot, chosen so the map tools can work.
- The original picture is never changed.

Self-test

Double-click **Run Rasta Self Test.bat** (or `py -3.14 "System Files\RPF_SELF_TEST.py"`). It checks the placement maths, the zoom rules and the TPKX converter (it converts a small test map and checks the result against the ArcGIS Earth-proven structure).